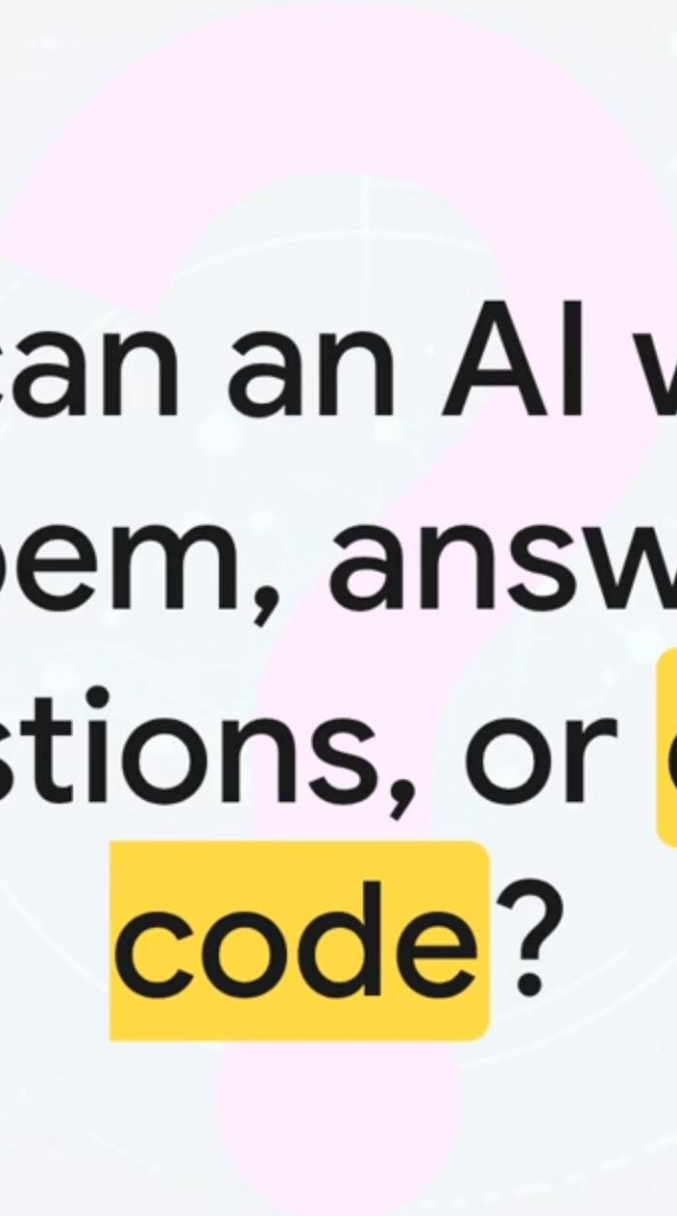




# Under the Hood: How Large Language Models *\*Really\** Work

From Statistical Prediction to Human-like Conversation



**How can an AI write a  
poem, answer  
questions, or even  
code?**



At their core, LLMs have one fundamental task: Predicting the next word in a sequence.

blue

The sun is shining and the sky is...

**blue**

bright

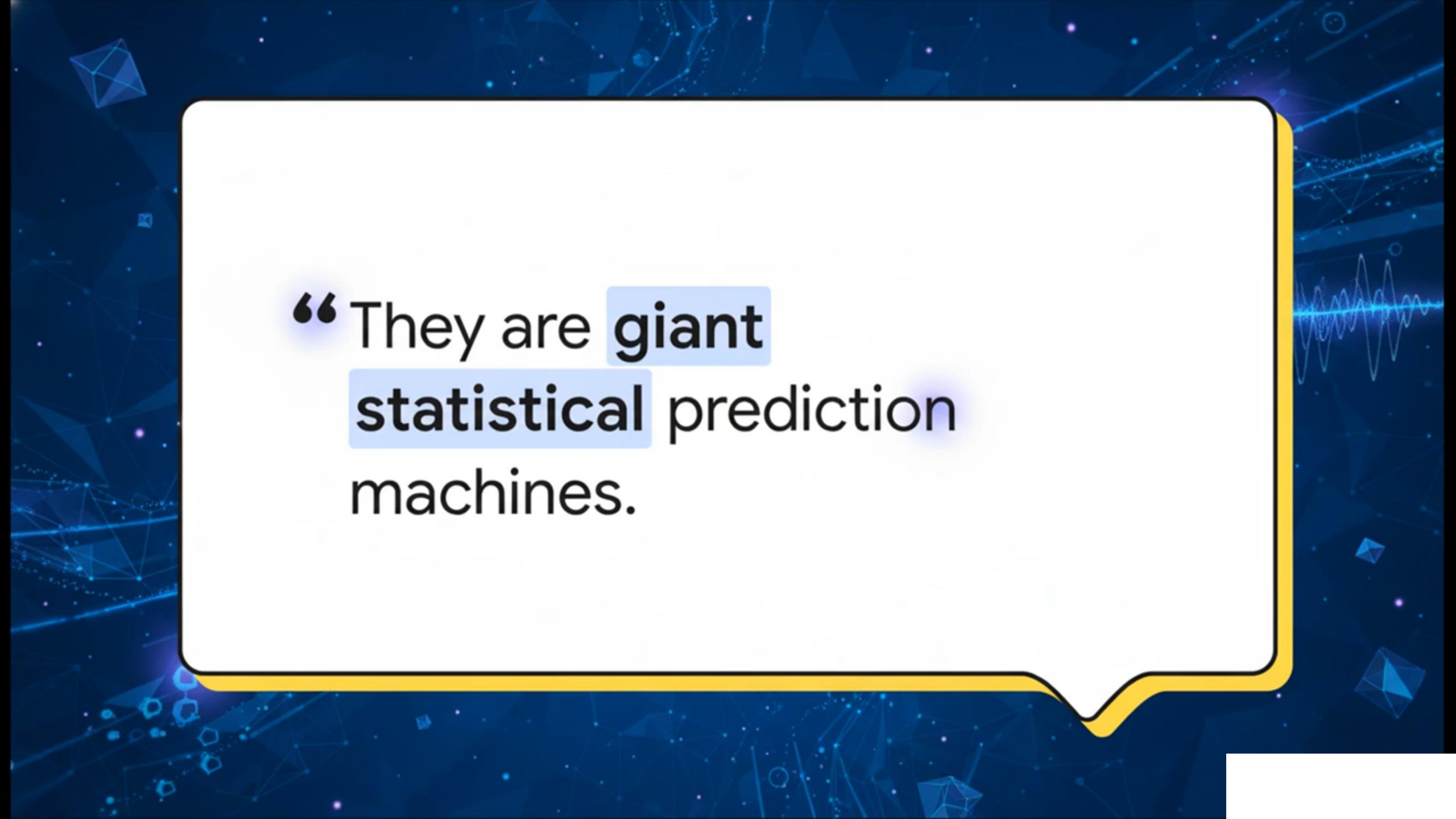
falling



# Large Language Model (LLM)

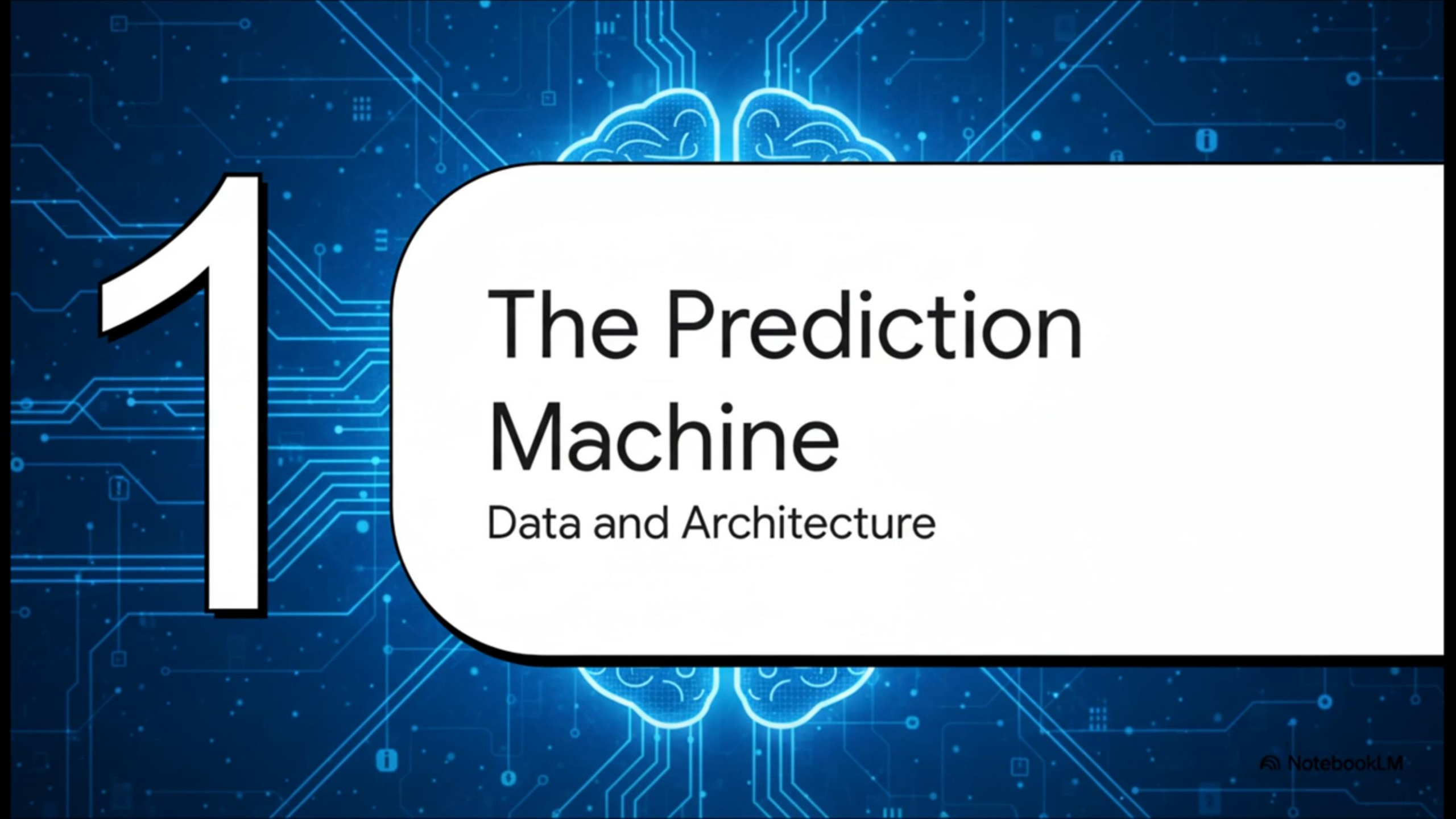
A neural network trained on vast text data to generate human-like language by predicting the next word.





“They are **giant**  
**statistical** prediction  
machines.

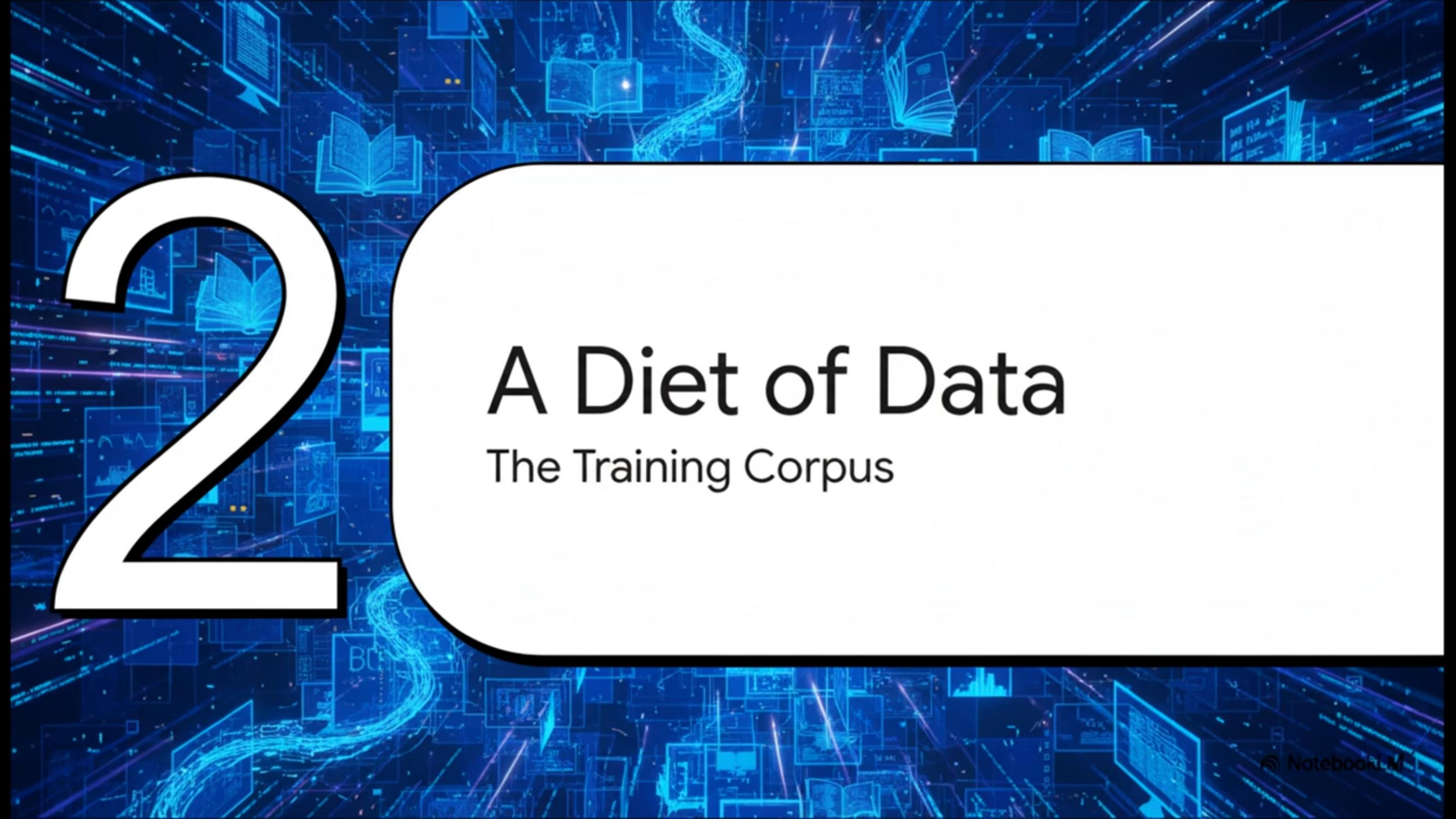




# 1

# The Prediction Machine

Data and Architecture



# 2

## A Diet of Data

The Training Corpus



## GPT-3 Data

- Common Crawl (60%)
- Curated Web Text (22%)
- Books (16%)
- Wikipedia (2%)

GPT-3 was trained on **300 billion** tokens from diverse text sources.





# Data Preparation

## Step 1: Collect

Gather massive text datasets from books, websites, and more.

## Step 2: Clean

Filter raw data to remove errors, duplicates, and unwanted content.

## Step 3: Tokenize

Break text into smaller pieces (tokens) the model can process.



unhappiness

The diagram illustrates the process of breaking a word into subword units. On the left, a large grey LEGO brick contains the word 'unhappiness'. Two white curved arrows point from this brick towards three smaller, colored LEGO bricks on the right. The first brick is blue and contains 'un', the second is yellow and contains 'happi', and the third is orange and contains 'ness'. This visualizes how a single word is decomposed into smaller, manageable units for processing.

un happi ness

Text is broken into smaller subword units. For example, 'unhappiness' becomes three separate tokens: ['un', 'happi', 'ness'].



# 3

## The Transformer Brain

Architecture of an LLM



# Embedding

A high-dimensional vector of numbers representing a token's semantic meaning. Similar words get similar vectors.







The vectors for 'King' and 'Queen' are close together, representing similar meanings.



The vectors for 'King' and 'Banana' are far apart, representing different meanings.

# Self-Attention Spotlight

The robot dropped the ball because it was too heavy.

The diagram illustrates the self-attention mechanism for the word "dropped". A central blue dot, representing the query vector for "dropped", is connected by lines to various words in the sentence. The lines are color-coded to represent attention weights: a thick red line connects to "robot", a thick blue line connects to "ball", and a thick green line connects to "was". Other words like "it", "because", "too", and "heavy" are connected by thinner lines, indicating lower attention weights. The background features a circuit-like pattern of blue lines and dots.



# 4

## Training for a Task

From Knowledge to Skill

# The Training Process



## 1. Pretraining

The model learns general language patterns from a vast, unlabeled text corpus.



## 2. Fine-Tuning

The model is trained on a smaller, labeled dataset for a specific task.



## 3. RLHF

Reinforcement Learning from Human Feedback aligns the model to be helpful.



## Pretraining



**Pretraining:** Builds general knowledge from a huge, unlabeled text corpus.



## Fine-Tuning



**Fine-Tuning:** Teaches a specific skill using a smaller, labeled dataset.

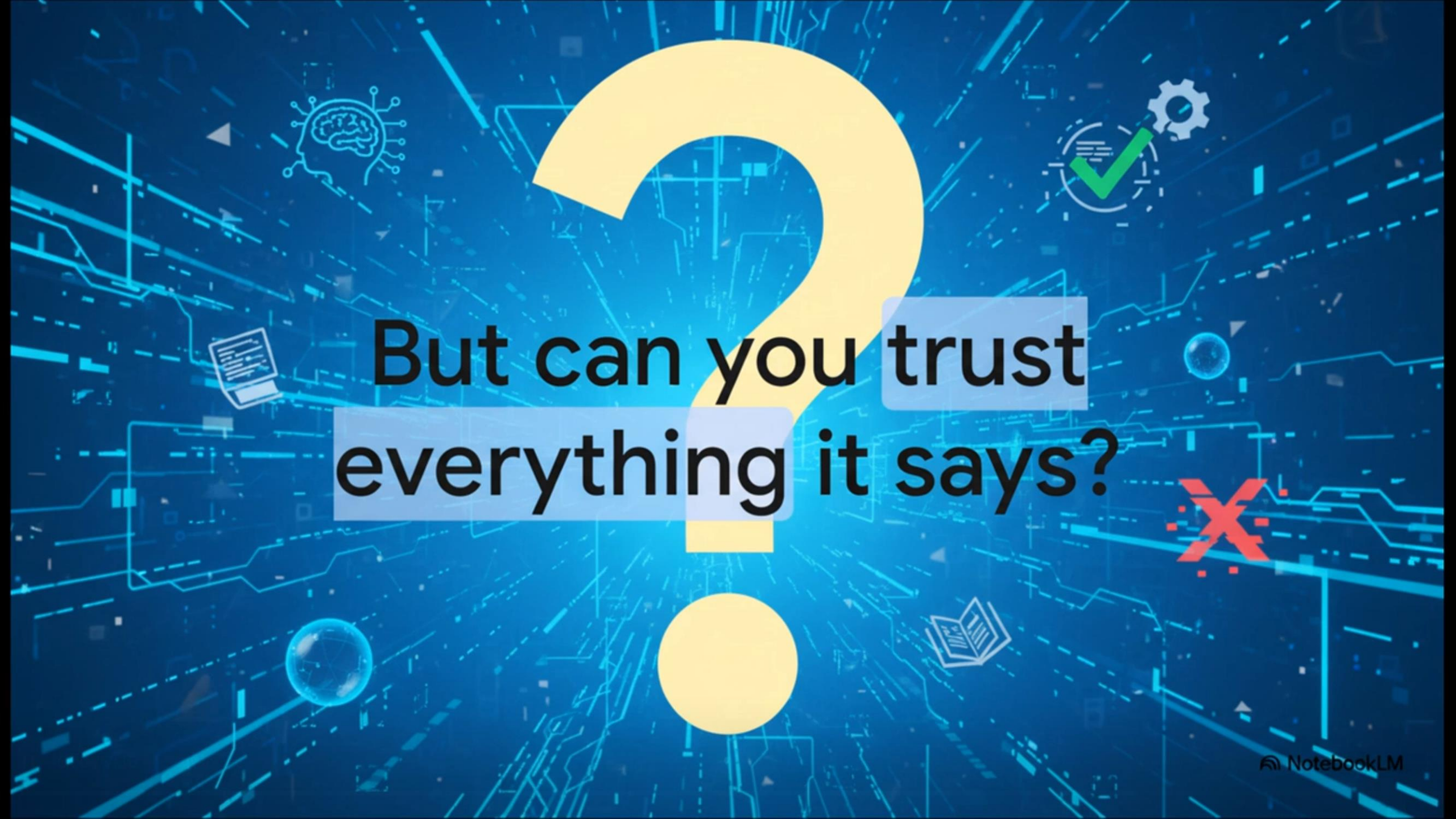


5

# LLM Challenges

Hallucinations and Bias





**But can you trust  
everything it says?**



# Hallucination

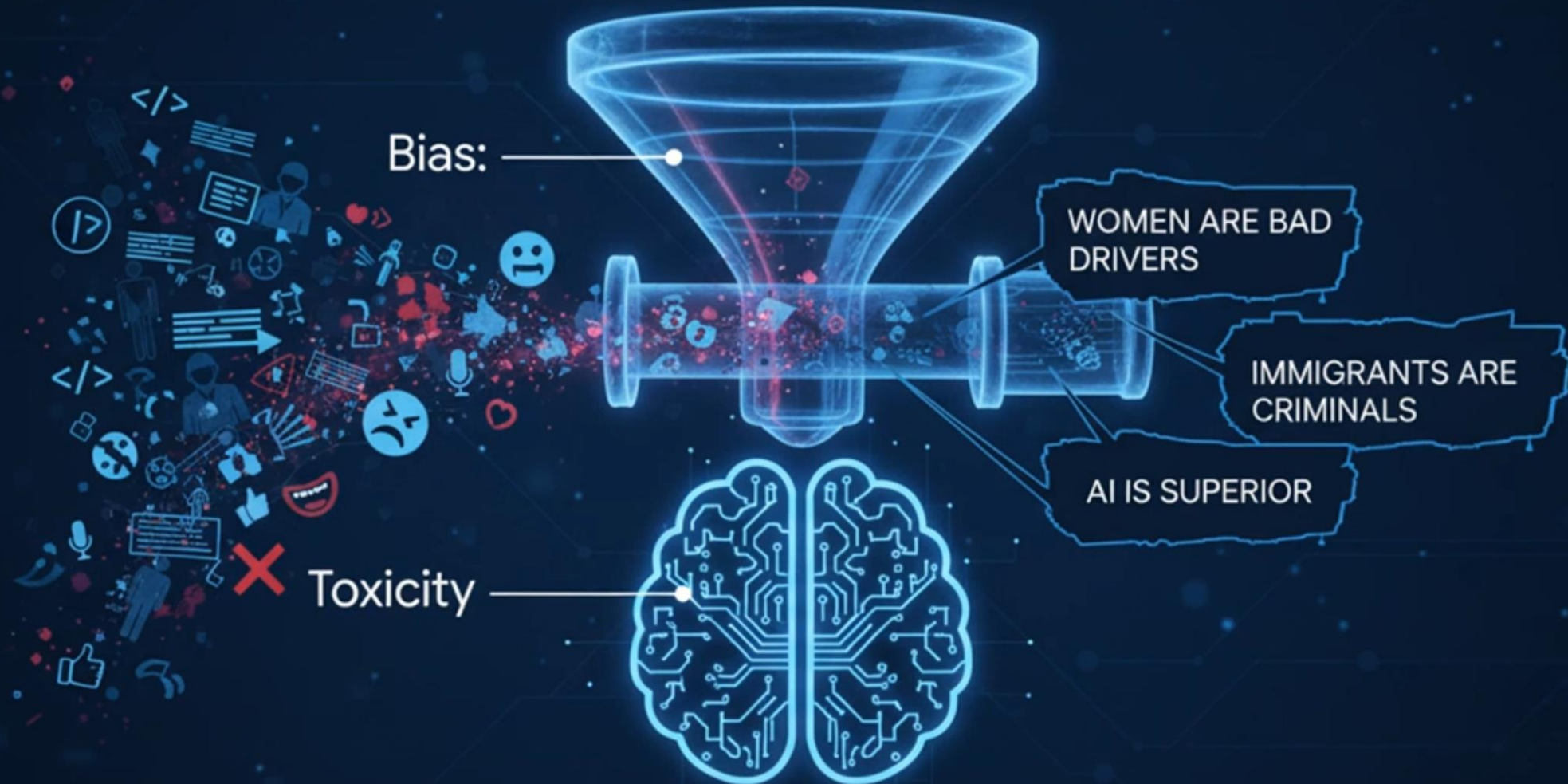
Generating plausible-sounding but factually incorrect or made-up content.





“It has no built-in **fact-checker**; it **makes things up** when uncertain.

# Data-Driven Challenges







How do we guide these  
powerful tools to work  
for everyone?





